

California Product Closure

Packet Assembly - Core Brief, Evidence Architecture, and Companion Federal Question

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Purpose of this packet

This packet asks a practical systems question: **Does the system remain legible and practically usable at the moment an adult resident is holding the actual object?** The proposal does not assume that California lacks waste management. It asks whether responsibility, information, access, capacity, and downstream accountability remain connected strongly enough to make lawful closure a realistic option.

A product lifecycle should include its closure. When a product is sold into California, the responsible producer should have a reasonably accessible pathway for repair, reuse, return, recycling, recovery, or lawful disposal. The goal is not to make every product recyclable. The goal is to make responsibility, information, funding, access, and capacity sufficiently clear that responsible closure is a realistic option.

The resident outcome test

After an intervention is implemented, could an ordinary adult resident actually do something differently, more easily, more safely, or more reliably than before?

For research purposes, outcomes are classified as: demonstrated; reported by an agency but not independently verified; partial; no demonstrated change; or unknown. A Grand Jury finding is preserved as a civic finding, not treated as a judicially established fact.

What the statewide Grand Jury scan contributes

Failure layer	Representative lesson
Information / legibility	Butte, Solano, Orange: information may exist but be confusing, outdated, inconsistent, or difficult to find.
Access / transportation	San Joaquin, Riverside, Calaveras: a service can exist without being practically usable for a resident without transportation, time, or convenient access.
Physical / technological capacity	Trinity, Kern, Orange: collection can exist while staffing, equipment, or processing capacity remains insufficient.
Funding / economics	Napa, Sonoma, Madera: market conditions, contracts, and operating costs can shift burdens without eliminating them.
Coordination	Mono, Humboldt, Kern, Riverside: fragmented jurisdictions can transfer coordination costs to residents and staff.

Failure layer	Representative lesson
Contracts / accountability	Orange, San Mateo, Lassen, Sacramento: downstream accountability and procurement structure affect trust and outcomes.
Implementation / follow-up	Madera, Humboldt, Riverside, Mono, Yolo: recommendation is not the same as implementation; follow-up reveals what changed.
Positive counterexamples	Marin, Lake, Yolo and parts of Del Norte: real success exists and should be treated as transferable evidence, not ignored.

Research synthesis

The evidence does not support the blanket claim that California waste management is simply dysfunctional. A more defensible conclusion is that California contains many functioning waste-management components, but their usability, accessibility, coordination, information quality, capacity, and downstream accountability vary substantially. The policy opportunity is to locate where the chain loses the resident and then apply successful existing solutions more systematically.

Two related but separate policy questions

California Product Closure: Can the adult resident holding the actual object identify, locate, afford, access, and complete a lawful closure pathway - and know who is responsible for the downstream outcome?

Federal / interstate / international material information: When the object crosses county, state, or national boundaries, does the information and responsibility necessary to manage it remain attached to the object or product record?

Key distinctions to preserve

Identification is not disposition. A material code, label, barcode, or product name does not by itself establish a lawful, accessible, functional end-of-life pathway.

Information existing somewhere is not the same as information being practically available. Data can be fragmented, outdated, or held by different organizations.

Collection is not closure. Moving material out of a residence does not demonstrate responsible downstream management.

Movement is not closure. A product being donated, resold, hauled, transferred, or exported may simply move responsibility to another actor.

Capacity matters. A webpage is not a disposal pathway unless sufficient physical, technological, financial, and institutional capacity exists behind it.

A verified absence is useful information. If no verified pathway exists, the system should be able to say so; the gap becomes a policy and capacity signal rather than a resident failure.

The information-layer concept

A future product-lifecycle information layer could let a resident scan a UPC/barcode or photograph an object, identify the likely product and producer, retrieve verified information about materials, repair/reuse options, safety, collection points, fees, authorized take-back, and downstream handling, and receive a transparent “no verified pathway found” result when evidence is insufficient. The technology is an interface, not the policy. **Local services remain local. The information architecture becomes interoperable.**

The “disappearance illusion”

Payment and transfer can make disposal feel finished even though the physical lifecycle continues. A more formal statement is: **Disposal is often experienced as disappearance because the physical and administrative consequences become invisible once the consumer has paid for the next handoff.** A system has performed a transfer, not necessarily a closure.

Capacity and investment

Complex products often require complex manufacturing technologies and similarly specialized end-of-life technologies. The policy question is not whether every product can be recycled. It is whether the system makes the necessary end-of-life capacity, cost, and responsibility visible enough to support realistic closure decisions.

What California is already doing

California already operates multiple producer-responsibility and product-stewardship programs, and in 2026 CalRecycle published SB 54 needs-assessment reports covering collection, processing, end markets, reuse/refill, and consumer education/access. The new state work explicitly examines resident knowledge, access, sorting, and the investments needed for collection and processing. This packet therefore asks for extension and integration of an existing policy direction rather than invention from scratch. Sources: CalRecycle, SB 54 Statewide Needs Assessment (2026); Consumer Education and Access Technical Report (2026).

Packet contents

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- 2. California Existing-Law Crosswalk** - comparison with current California EPR/product-stewardship models.
- 3. Santa Cruz County & California Product Closure - Revised Access Gap** - local access and system comparison.
- 4. WaveStorm Test Case** - concrete product example.
- 5. Federal Material Information Companion** - separate federal/interstate/international research question.
- 6. Evidence / library layer** - Grand Jury reports, responses, follow-ups, CalRecycle sources, and field notes retained as a deeper research library.

Source notes for this assembly

This working packet incorporates current official CalRecycle material and the statewide Grand Jury research reviewed during the scan. The Grand Jury evidence is treated as a body of local civic observations: findings, recommendations, agency responses, follow-ups, and outcomes are kept distinct. Where an outcome could not be independently verified, the matrix says so rather than implying certainty.

Primary current state references

CalRecycle, "SB 54 Statewide Needs Assessment" (current page, accessed September 2026): the assessment covers collection, processing, end markets, reuse/refill, consumer education/access, and needed investments. CalRecycle, "Consumer Education and Access; A Plastic Pollution Prevention and Packaging Producer Responsibility Act Needs Assessment Technical Report," publication date February 18, 2026. CalRecycle reports the study surveyed 2,297 adult California residents and assessed knowledge, access, sorting, and participation. These materials support the packet's distinction between formal program existence and practical resident usability.

Statewide civic-observation reference

California Civil Grand Jury Association / statewide Grand Jury archive topic materials used during the scan. Final publication should cite each original county report and response individually for claims drawn from that report, rather than relying solely on topic-index counts.

California Product Closure Proposal

A working framework for accessible, proportional end-of-life responsibility

Core idea

A product lifecycle should include its closure. When a product is sold into California, the responsible producer should have a reasonably accessible pathway for repair, reuse, return, recycling, recovery, or lawful disposal when that product reaches the end of its useful life.

Why this matters

California already uses extended producer responsibility (EPR) and product-stewardship approaches for selected product categories. The question is whether the underlying principle should be made more systematic so that ordinary consumers are not left to solve end-of-life problems that were built into the products they were sold.

Modern commerce makes acquisition highly visible and disposal comparatively opaque. A consumer may know where an item was purchased while having no practical answer to the question: “Where does this go when I am finished with it?” That uncertainty can push useful objects into free boxes, illegal dumping, ordinary trash, or expensive specialized disposal.

Policy objective

Reduce the end-of-life mystery in consumer commerce. The goal is not to make every material recyclable or to impose one disposal method on every product. The goal is to make responsibility, information, funding, and access sufficiently clear that responsible closure is a realistic option.

Proposed framework

Element	Proposed standard
1. Identify responsibility	Identify the manufacturer, brand owner, importer, or other responsible producer.
2. Identify the pathway	Provide a repair, reuse, take-back, recycling, recovery, or lawful-disposal pathway appropriate to the product.
3. Make access practical	Provide geographically reasonable access, including through authorized retailers, repair businesses, reuse organizations, transfer stations, or other partners.
4. Make funding proportional	Use producer funding, point-of-sale fees, deposits, stewardship organizations, or comparable mechanisms so end-of-life costs are not unexpectedly shifted to the final owner.
5. Preserve reuse	Treat repair, resale, donation, refurbishment, free exchange, and other legitimate second-life pathways as valuable outcomes.
6. Address orphan products	Provide a mechanism for products whose original producer has disappeared or cannot reasonably provide service.
7. Include Internet sales	Products sold into California through electronic commerce should not escape applicable responsibility.

Element	Proposed standard
8. Protect small businesses	Do not require small retailers to independently finance or operate specialized waste facilities; allow participation in shared producer-funded systems.
9. Measure results	Track collection, reuse, recycling, disposal, program cost, consumer access, illegal dumping, environmental outcomes, and economic effects.
10. Allow evolution	Create a transparent process for adding product categories as evidence identifies significant end-of-life gaps.

Proportionality principle

The cost of responsible closure should not be so disproportionate to the ordinary value of a product that abandonment, illegal disposal, or an informal free box becomes the economically rational choice for the consumer.

What this proposal does not require

It does not require every retailer to accept every product; does not require every material to be recycled; does not prohibit durable plastic products; and does not assume that government alone should operate the system. It creates a framework in which producers, retailers, waste companies, recyclers, repair businesses, nonprofits, and local governments can participate.

Initial pilot question

California could examine difficult-to-manage consumer products for which ordinary owners lack a clear, affordable end-of-life route—such as selected composite plastics, foam products, sporting and recreational equipment, and other categories identified through CalRecycle analysis.

Six-question consumer test

1. Who made it? 2. Who sold it? 3. What is it made of? 4. Where can I repair or reuse it? 5. Where can I return it? 6. Who pays for what happens next?

If a system cannot answer questions 5 and 6, the product lifecycle is incomplete.

Request to policymakers

Please evaluate existing California product-stewardship and EPR statutes and determine what amendments, pilot programs, regulations, or new legislation could create a broader, proportional end-of-life access framework.

Working note

This is a citizen policy proposal, not legal advice. Existing California programs and statutes should be verified against current official sources before legislative drafting.

California Existing-Law Crosswalk

What California already does—and where the proposed Product Closure framework could extend it

Starting point

California is not starting from zero. The state already operates and authorizes multiple extended producer responsibility and product-stewardship programs. The policy question is therefore one of scope, consistency, access, financing, and future expansion.

Crosswalk

Existing California model	What it demonstrates	Potential adjustment / lesson
Electronic waste	Point-of-sale fees and approved collection/recycling mechanisms can finance end-of-life management for covered electronics.	Use proportional financing and authorized collection networks as a model for additional difficult products.
Batteries / battery-embedded products	California has established producer/retailer obligations and fee mechanisms for certain battery-related products.	Consider product-specific risk and cost while retaining a general framework for identifying future categories.
Paint	Producer responsibility can support statewide collection and recycling infrastructure.	Demonstrates that consumers can be given a recognizable return pathway rather than being left to discover disposal options independently.
Mattresses	A dedicated stewardship system can fund collection and recycling for a bulky product category.	Useful model for large objects that are inconvenient to transport through ordinary curbside systems.
Carpet	A product category can have dedicated end-of-life requirements and funding.	Demonstrates that difficult composite material streams can receive specialized treatment.
Textiles	California can establish statewide producer responsibility for a large, diffuse consumer-product stream.	Supports the concept of phased expansion rather than requiring every product to be covered immediately.
Packaging	Producer responsibility can address products entering the waste stream at very large scale.	Provides a model for financing and measurable statewide outcomes.
Beverage containers	California already uses a deposit/refund mechanism that changes behavior at the point of purchase and return.	Shows another way to connect acquisition with eventual return.

What appears to be missing

The existing programs are generally product-specific. They do not create a universal consumer right to return every ordinary consumer product to its seller or producer. That leaves many products in a gray zone: the owner may have a lawful disposal obligation but no convenient, recognizable producer-funded pathway.

Questions for legislative counsel and CalRecycle

1. Which existing statutes could authorize a general product-stewardship framework without duplicating existing programs?
2. Could California create a standing process for designating additional product categories when an end-of-life gap is demonstrated?
3. What proportional financing mechanisms are legally and administratively practical for lower-value products?
4. How should Internet sales and products from manufacturers outside California be handled?
5. How should orphan and historic products be managed?
6. What geographic access standard should apply, especially in rural communities?
7. How can small retailers participate without becoming financially liable for specialized waste processing?
8. What existing recycling, repair, reuse, transfer, and waste infrastructure can be incorporated?
9. What metrics would demonstrate that a program reduces illegal dumping, landfill disposal, or unreasonable consumer burden?
10. What products should be selected for pilot programs first?

Policy adjustment rather than policy replacement

The proposal should be presented as an evolution of California's existing approach: retain specialized programs where they work, establish a general framework for identifying uncovered end-of-life gaps, and create a transparent pathway for adding products when evidence supports intervention.

The proportionality test

A responsible end-of-life pathway should be economically and geographically realistic for ordinary owners. The objective is not necessarily zero cost to consumers; it is to prevent a situation in which the formal responsible pathway costs so much more than the product's ordinary value that abandonment becomes the predictable outcome.

The “mystery” in secular terms

Modern commerce can be highly sophisticated at producing and selling things while remaining opaque about what happens to them afterward. This is an **end-of-life knowledge gap**: responsibility, information, and physical material flows are separated among manufacturers, retailers, consumers, haulers, recyclers, governments, and disposal facilities.

Policy cannot eliminate every complexity. It can, however, **make the mystery smaller** by connecting product design, sale, ownership, return, financing, and final management into a more visible lifecycle.

Lifecycle endpoint principle

Every material system should have a planned end point. If we create something, we should eventually be able to answer what happens when its useful life is over. A responsible system does not stop at production or first use; it includes closure.

Sources to verify during formal drafting

CalRecycle's current Extended Producer Responsibility and product-stewardship pages; current California statutes and regulations for covered product categories; Senate Environmental Quality Committee; Assembly Natural Resources Committee; relevant Department of Toxic Substances Control and local solid-waste authorities. This crosswalk is a working policy document and should not substitute for legislative counsel's review.

Santa Cruz County & California Product Closure

A policy comparison focused on access, responsibility, and the end-of-life knowledge gap

Prepared by Paul Statchen

Santa Cruz, California · 2026

Purpose

Santa Cruz already has substantial recycling, waste-reduction, household-hazardous-waste, bulky-item, and resource-recovery infrastructure. California also has established product-stewardship and extended-producer-responsibility (EPR) programs. The question here is not whether government is doing anything. It is whether the existing system provides an understandable, reasonably accessible, and proportionate closure pathway for products that do not fit cleanly into an existing stream.

Existing system compared with the proposed framework

CITY OF SANTA CRUZ

What exists: The City provides recycling and resource-recovery services, including the facility at 605 Dimeo Lane, plus programs for particular materials, bulky items, organics, and household hazardous waste.

Policy question: Strength: a real local service network. Limitation to examine: residents still have to determine which program or facility accepts a particular item.

SANTA CRUZ COUNTY

What exists: The County provides solid-waste and recycling information, disposal facilities, transfer-station options, household hazardous-waste services, and information about other facilities and disposal requirements.

Policy question: Strength: countywide infrastructure and information. Limitation to examine: different materials, facilities, jurisdictions, hours, fees, and transportation requirements can make the pathway difficult to navigate.

STATE OF CALIFORNIA

What exists: CalRecycle operates product-stewardship/EPR programs for specific categories including textiles, beverage containers, batteries, packaging, pharmaceuticals and sharps, mattresses, carpet, and paint.

Policy question: Strength: California already places end-of-life responsibility on producers for selected product categories. Limitation to examine: the principle remains category-specific.

PROPOSED PRODUCT CLOSURE FRAMEWORK

What exists: Extend the underlying responsibility principle to additional products where appropriate. A product entering the marketplace would have a planned pathway for repair, reuse, return, recycling, recovery, or lawful disposal.

Policy question: Goal: make the closure pathway part of the product lifecycle rather than leaving the final consumer to discover and fund the solution alone.

The access problem

The existence of a recycling or disposal program does not necessarily mean that a resident can readily identify an acceptable, affordable, and practical destination for a particular object. Official information can require the resident to identify the material category, determine which facility accepts it, determine whether that facility serves the resident's jurisdiction, determine hours and transportation rules, and determine whether a fee applies.

System capacity is not the same thing as user access

A system can have facilities, phone numbers, maps, programs, and educational materials while an individual still encounters a chain of referrals: the first location does not accept the item and directs the person elsewhere; the next location may have a different restriction or cost. For a person without a vehicle, without money for a private hauler, or without enough information to classify the material correctly, the practical pathway can become difficult even when a legal disposal pathway technically exists.

The affordability and equity question

This paper does not assume that private waste haulers are acting improperly. Transportation, labor, equipment, insurance, permits, disposal fees, and regulatory compliance all have real costs. The policy question is whether the resulting cost or logistical burden creates a disproportionate barrier for residents with fewer financial resources.

The California connection

California's EPR framework already recognizes producer responsibility for specified product categories. The proposed framework does not reject that system. It asks whether the same design logic can be applied more broadly: if a product enters the California marketplace, its lifecycle should include a reasonably accessible and proportionate method of closure.

A practical policy test

1. **Responsibility:** Who is responsible for the product at end of life?
2. **Pathway:** Is there a defined route for repair, reuse, return, recycling, recovery, or lawful disposal?
3. **Accessibility:** Can an ordinary resident find the correct destination without an unreasonable chain of referrals?
4. **Affordability:** Are costs proportionate to the product and the environmental/public consequences of improper disposal?
5. **Jurisdiction:** Does the pathway work across city, county, and state boundaries where appropriate?
6. **Accountability:** Can the public and government determine whether the material is actually managed as intended?

A local pilot question

Santa Cruz County could examine a small number of difficult, commonly encountered products as a demonstration project. For each product, the pilot could document where residents are currently told to take it; whether the destination accepts it; transportation requirements; fees; jurisdiction restrictions; reuse or repair possibilities; final disposition; and who ultimately bears the cost.

The central distinction

Existing policy asks: “Where does this material belong?”

The proposed framework adds: “Who is responsible for making sure a workable place exists for it to go?”

That distinction preserves recycling education, local facilities, private haulers, producer responsibility, reuse, and environmental regulation while asking whether the entire pathway is understandable and accessible from the resident's point of view.

Sources reviewed

City of Santa Cruz recycling and resource-recovery information; Santa Cruz County Recycling & Solid Waste information and disposal-facility information; CalRecycle Product Stewardship and Extended Producer Responsibility information; and CalRecycle materials concerning effective EPR program design.

This is a policy working paper, not a legal conclusion. Its purpose is to support a constructive discussion about how existing recycling, waste-management, and producer-responsibility systems might be made easier to navigate and more equitable in practice.

The WaveStorm Test Case

A real-world example of the end-of-life access problem

The object

A used WaveStorm-style soft-top surfboard was found through community reuse. It remains potentially useful, so the immediate decision is not disposal. The larger question is what happens later, when the board can no longer be safely or usefully repaired or passed to another person.

The community-reuse observation

A free box on the street can be a responsible act: someone has tried to sell, donate, repair, or otherwise transfer an item and is offering it to another person rather than sending it directly to disposal. But the free box can also reveal a missing formal pathway. If the original owner had difficulty finding a legitimate destination, the next owner may inherit the same problem.

The question

If a person in Santa Cruz owns a foam-and-plastic surfboard that has reached the end of its useful life, can they identify a local, lawful, reasonably affordable pathway for repair, reuse, material recovery, or disposal? If the answer is unclear, who should be responsible for creating that answer?

Why the seller/producer is a logical starting point

The manufacturer and seller know the product's materials, construction, supply chain, intended use, and likely failure modes better than an ordinary consumer does. They are therefore in a better position to identify appropriate end-of-life partners and to incorporate the expected cost into the product system.

What a practical take-back system could look like

Consumer → authorized seller or collection partner → consolidated products → repair/reuse where feasible → material recovery where feasible → controlled disposal for residual material.

The retailer would not necessarily process the board. The retailer could simply serve as an identified return point, while the producer-funded network handles transportation and downstream management.

Why proportional cost matters

A low- or moderate-value consumer product should not require the owner to pay several times the product's ordinary value merely to dispose of it responsibly. The system should anticipate this mismatch and finance closure before the product becomes an orphaned object.

The policy test

A consumer should be able to ask: "I bought or acquired this product in California. It is now finished. Where does it go?" The answer should be identifiable without specialized knowledge, extraordinary travel, or an

economically irrational disposal fee.

What should happen to this particular board?

For the board while it remains usable: repair and second-life use should come first. Before attempting a final disposal route, confirm the condition of the foam, deck, rails, bottom, fins, leash attachment, and any structural damage. At end of life, contact the manufacturer/brand, local surf businesses, reuse organizations, and the appropriate local waste authority to determine the current accepted pathway. Do not assume a general rigid-plastics recycling program accepts a composite foam surfboard.

From one board to a statewide test

The point is not that one surfboard requires a special law. The point is that a single ordinary object can expose a general systems question: why can California make it easy to buy an object without making its eventual destination equally legible?

Citizen proposal

Use difficult-to-manage products such as foam recreational equipment as pilot cases for a broader Product Closure framework. Ask manufacturers, retailers, waste operators, local governments, and CalRecycle to jointly map the actual material pathway and identify the least-cost responsible closure option.

Working note

This document is a policy case study, not a claim that a particular company currently accepts this board for recycling. Acceptance must be confirmed directly with the relevant business or public agency.

Federal Material Information Companion

A separate research question for imports, interstate movement, ports, and downstream traceability

Purpose

This companion does not accuse ports, carriers, states, or foreign countries of failing to track waste. The narrower question is what product and material information is already captured at import, manufacture, distribution, interstate transportation, and regulated waste shipment stages - and what additional information would be necessary to connect origin information to responsible end-of-life management.

When material crosses a jurisdictional boundary, does the information and responsibility necessary to manage it remain attached to the material or product record - or does only the material move?

Three levels

Local: resident holding the object; local collection, reuse, repair, transfer, processing, disposal.

Interstate: trucking, rail, distribution, processors, disposal facilities, secondary material markets.

International: ocean/air cargo, ports, import/export, cross-border processing, international material markets and applicable transboundary controls.

What should be investigated

What identifiers are already present in commercial and regulatory records: product IDs, manufacturer, importer, material category, shipment, destination facility, and responsible entity?

Which information can be linked across systems without creating a new universal tracking mandate?

Where do information breaks occur when products are resold, repackaged, consolidated, exported, or transferred between jurisdictions?

Which regulated waste streams already demonstrate shipment-level traceability, and what can be learned without assuming those rules apply to ordinary consumer goods?

Can a common product-data standard make local and interstate closure information interoperable while leaving operations decentralized?

Important caution about ISO 14001

ISO 14001 is a voluntary environmental management system standard. It should be treated as a possible source of organizational environmental-management information, not as evidence that every manufacturer has complete product-level end-of-life traceability. The federal question should distinguish documented information systems from assumptions about corporate practice.

Cross-jurisdictional displacement

A product may move from local collection to another county, state, or country because of capacity, transportation cost, processing economics, regulatory requirements, or end-market conditions. The research question is whether information about responsibility and final disposition becomes harder to determine as the number of transfers increases. We should test this empirically rather than assume an exponential effect.

Why this belongs beside Product Closure

The California proposal asks whether the resident can successfully close the lifecycle. The federal companion asks whether responsibility and information can remain connected to the product as it crosses jurisdictional boundaries. The two questions are related but should remain separate so the California proposal stays concrete and actionable.

Current federal reference points

EPA maintains requirements and information systems for certain transboundary hazardous-waste shipments, including exporter/importer, transporter, and receiving-facility information, while international controls differ by waste type and treaty status. These regulated systems provide precedents for studying traceability without implying universal product tracking. Primary reference: U.S. EPA, Information About Transboundary Shipments of Hazardous Wastes; International Agreements for Transboundary Shipments of Hazardous Waste.